04-family-sweep: results

Pair-46 factor-base campaign, item 04. Run 2026-10-15T15:37Z, seed 46004, protocol `PROTOCOL.md`.

The protocol was frozen before data; its sha256 is in `PROTOCOL.sha256`. Raw per-curve rows are in `curves.csv`

(500 curves); tests and the add-on are in `results.json`; the log is `run.log`.

Verdict: null. On all three preregistered metrics, the family p = 5t^2+19t+19 is within

the control distribution of both nulls, and all three positive controls were detected. There is no

family-level factor-base advantage at 20-30 bits.

Primary table (all bits pooled, stratified by bit length; Bonferroni α = 0.05/6)

metric	F mean	null	null mean	diff [95% CI]	ratio [95% CI]	p (perm.)	positive control detected?	verdict
M1 normalized Semaev m=2 yield	0.9928	N1 D=-19 non-family	1.0087	-0.0160 [-0.0364, +0.0045]	0.984 [0.964, 1.005]	0.144	yes (PC/N2 1.040 [1.020, 1.061], p=0.0004)	within control distribution
M1	0.9928	N2 random	1.0078	-0.0150 [-0.0353, +0.0047]	0.985 [0.965, 1.005]	0.153	yes	within control distribution
M2 relation rate	0.6293	N1	0.6353	-0.0060 [-0.0159, +0.0035]	0.991 [0.975, 1.006]	0.262	yes (PC/N2 +0.0194 [+0.0098, +0.0297], p=0.0005)	within control distribution
M2	0.6293	N2	0.6360	-0.0066 [-0.0167, +0.0033]	0.990 [0.974, 1.005]	0.227	yes	within control distribution
M3 small-x yield	1.00076	N1	0.99971	+0.00105 [-0.00167, +0.00365]	1.0011 [0.9983, 1.0037]	0.472	yes (PC/N2 +0.0060 [+0.0039, +0.0080], p=0.0001)	within control distribution
M3	1.00076	N2	1.00039	+0.00037 [-0.00211, +0.00293]	1.0004 [0.9979, 1.0029]	0.797	yes	within control distribution

n: F=100 (every family member with p and N=p+1-t prime in 20..30 bits), N1=200, N2=200, PC=100; T=120 random targets per curve.

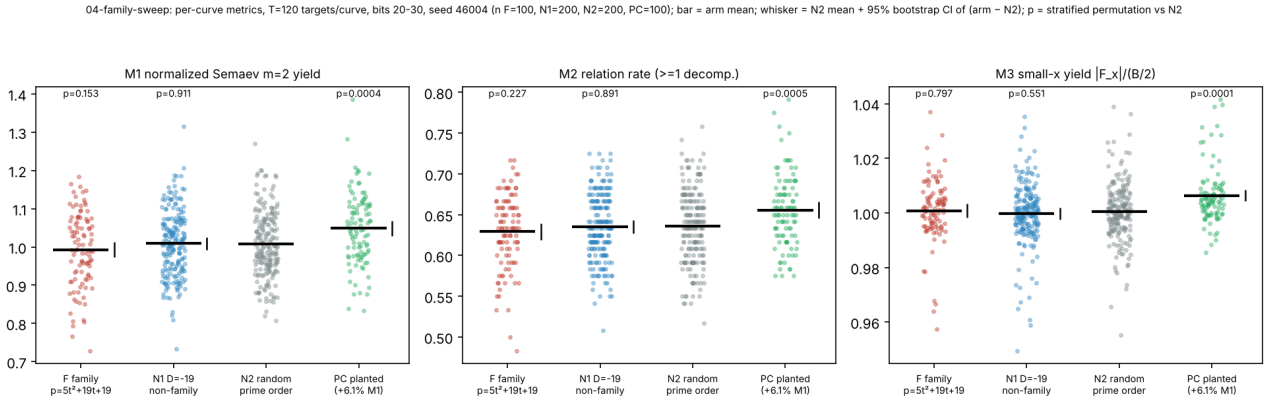
Secondary: N1 vs N2 (does D=-19 CM matter?) is within the control distribution on all three metrics

(M1 diff +0.0009 [-0.015, +0.017], p=0.91; M2 p=0.89; M3 p=0.55).

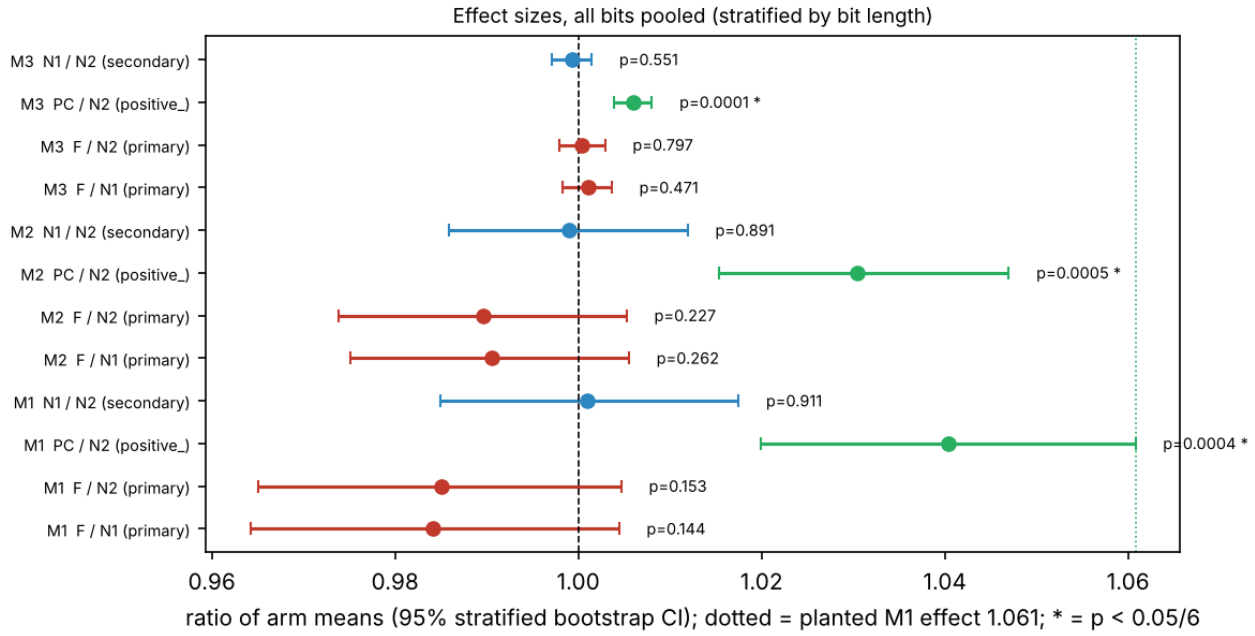
Per-band results are descriptive only (all of them are in `results.json`). The smallest p among them is

for M1 F vs N2 in the 25-27 bit band: diff -0.041 [-0.086, -0.0004], p=0.059. That p is uncorrected, and the effect is in the *unfavourable*

direction. No per-band test passes α/6.



Per-curve metrics by arm. Bar = arm mean; whisker = N2 mean plus the 95% bootstrap CI of (arm - N2); p = stratified permutation test against N2. 500 curves, T=120, seed 46004.



Effect sizes as ratios of arm means with 95% stratified-bootstrap CIs. The dotted line is the planted M1 effect 1.061. * marks p < 0.05/6.

Positive-control note

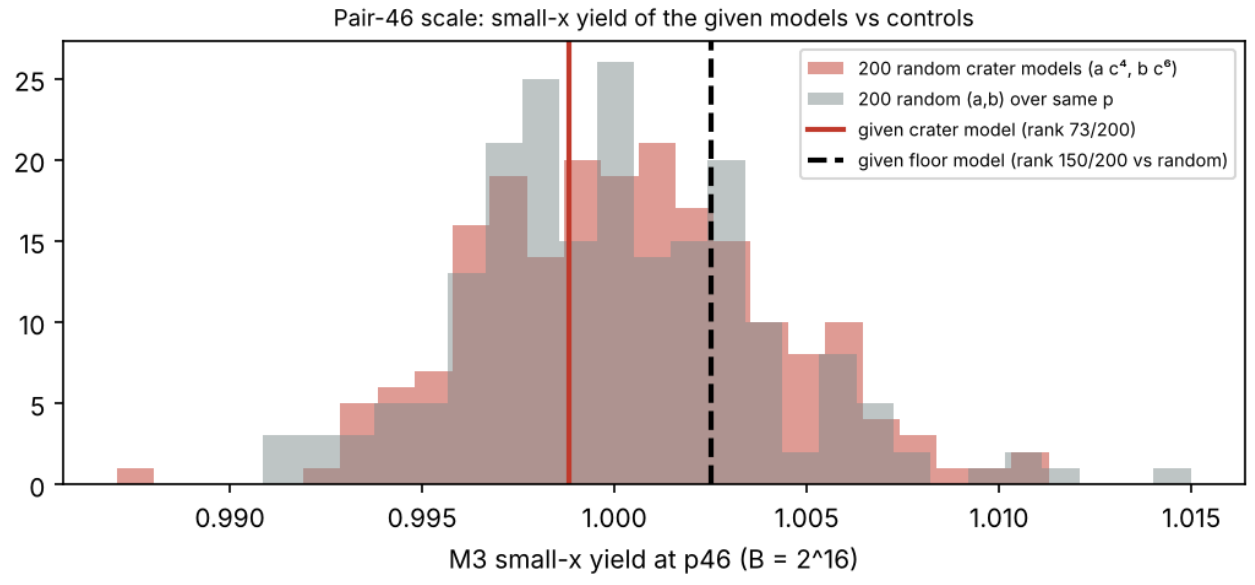
The planted M1 effect was a 3% wider membership bound (ratio 1.03^2 = 1.061). The PC also used best-of-2 model selection, which raises the normalizing |F_x| by about 0.6% (the measured PC M3 ratio is 1.006), so the expected M1 ratio is about 1.061/1.012 = 1.048. The observed ratio is 1.040 [1.020, 1.061], consistent with that. At this sample size the detector resolves effects of roughly +-2% in M1, +-0.01 in M2 and +-0.0025 in M3 (the CI half-widths).

Pair-46-scale add-on (M3 only, B = 2^16)

model	M3	rank among 200 crater models	rank among 200 random (a,b)
given crater	0.99881	73/200	82/200
given floor	1.00250	n/a	150/200

Control medians: crater models 1.00031 [0.99365, 1.00812], random 0.99973 [0.99265, 1.00922].

Both given models are within the control distribution.



Pair-46: small-x yield of the given crater and floor models against 200 random crater models and 200 random curves.

Exact vs empirical

• Exact: t^2 - 4p = -19(t+2)^2 for p = 5t^2+19t+19 (algebra; pair 46 is checked in the run). The crater has j = -96^3 and

#E = N, and the floor has #E = N and j != -96^3 (checked by computation). The S3 summation-polynomial self-test passed.

For D = -19 the unit group is {±1}, so Aut(E) = {±1} and the small-x factor base gets no extra orbit

compression (standard theory, not computed here).

• Empirical: every comparison in the tables. These are statements about 20-30-bit curves, the small-x factor base,

m=2 Semaev decomposition and B = ceil(sqrt(2N)).

What was cut / scope

• Floor curves (End = Z[pi], conductor t+2) of family members were not generated, because that needs a volcano descent. The family

arm is crater curves only.

• Semaev M1/M2 at 53 bits (pair 46 itself) was not run, because it needs about 5e7 root-finds per target. Only M3 was measured at pair-46 scale.

• Only m=2 (S3) was run. m >= 3 decompositions, and factor bases other than small-x, are not covered.

• Scope: "within the null on these metrics at 20-30 bits" does not mean that no family property helps, and it

does not establish anything about 53-bit behaviour beyond the M3 add-on.

Run facts

The pre-flight estimate was about 850 s of one core. Actual: 612 s user, 749 s wall (`stdout.txt`), pinned to 1 core (taskset -c 3).

All 100 family members and every planned control ran; none failed. Python only lays out the figures and the PDF; every number

comes from the Rust binary.

Reproduce

```
cd research/pair46_factor_bases_20261015/04-family-sweep
cargo build --release && taskset -c 3 target/release/family_sweep . 120 20 30 9999 2000
python3 plot.py . && python3 make_pdf.py .
```